

Fractional Expression

A fractional expression is a fraction whose numerator and denominator are algebraic expressions.

A special case is rational expressions where the numerator and denominator are polynomials.

Fractional expression e.g.: $\frac{12-y}{\sqrt[3]{x}-4}$

Rational expression e.g.: $\frac{x^2+5x-3y}{y^3-x^2}$

Problems involving rational expressions will usually ask you to add, subtract, multiply, or divide them and/or then simplify by cancelling common factors from num. and denom.

$$\frac{12+r-r^2}{r^3+3r^2} \rightarrow \text{simplify}$$

$$\frac{36}{22} = \frac{\cancel{2} \cdot 18}{\cancel{2} \cdot 11} = \frac{18}{11}$$

$$\frac{12+r-r^2}{r^3+3r^2} = \frac{12+r-r^2}{r^2(r+3)}$$

$$= \frac{-r^2 + r + 12}{r^2(r+3)}$$

$$= \frac{-r^2 + 4r - 3r + 12}{r^2(r+3)}$$

$$= \frac{(-r-3)(r-4)}{r^2(r+3)}$$

$$\frac{6}{x(3x-2)} + \frac{5}{3x-2} - \frac{2}{x^2}$$

$$a=-1, b=1, c=12$$

p	q
-1	12
-2	6
-3	4

Combine and simplify

$$\frac{6}{x(3x-2)} + \frac{5}{3x-2} - \frac{2}{x^2}$$

↓ at least one x, at least one 3x-2
 ↓ at least one 3x-2
 ↓ at least two x's

$$LCD = x^2(3x-2)$$

$$\frac{a^3-8}{a^2-4} \div \frac{a}{a^3+8} \rightarrow \text{Combine and simplify}$$

diff. of cubes
sum of cubes

$$\frac{a^3-8}{a^2-4} \cdot \frac{a^3+8}{a} = \frac{(a^3-8)(a^3+8)}{(a^2-4)a}$$

diff. of squares

$$a^2-4 = a^2-2^2 = (a-2)(a+2)$$

$$a^3-8 = a^3-2^3 = (a-2)(a^2+2a+4)$$

$$a^3+8 = a^3+2^3 = (a+2)(a^2-2a+4)$$

$$\frac{x^2 - 4}{3x^2 - 5x + 2} \cdot \frac{9x^4 - 6x^3 + 4x^2}{27x^4 + 8x} \rightarrow \text{Combine and simplify}$$

$$\frac{(9x^2 - 4) \cdot (9x^4 - 6x^3 + 4x^2)}{(3x^2 - 5x + 2) \cdot (27x^4 + 8x)}$$

$$9x^2 - 4 = (3x)^2 - 2^2 = (3x - 2)(3x + 2)$$

$$9x^4 - 6x^3 + 4x^2 = x^2(9x^2 - 6x + 4)$$

$\begin{array}{c} \nearrow 36 \nwarrow q \\ p \quad -6 \quad \searrow r \\ \rightarrow + \end{array}$

$$\begin{aligned} 3x^2 - 5x + 2 &= 3x^2 - 3x - 2x + 2 \\ &= 3x(x - 1) - 2(x - 1) \\ &= (3x - 2)(x - 1) \end{aligned}$$

$$\begin{array}{c} x \nwarrow \\ \nearrow 6 \quad \nwarrow q \\ p \quad -5 \quad \searrow r \\ \rightarrow + \end{array}$$

$$\begin{aligned} 27x^4 + 8x &= x(27x^3 + 8) \\ &= x(3x + 2)(9x^2 - 6x + 4) \end{aligned}$$

$p = -3, q = -2$

$$\frac{x}{x - 1}$$

$$\frac{\sqrt{x+h} - \sqrt{x}}{h} \rightarrow \text{rationalize the numerator}$$

$$\frac{\sqrt{x+h} - \sqrt{x}}{h}$$

↓
use conjugate
(ultimately from diff.
of squares)

num. became just $x+h-x$

once everything cancels

$$= \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$\frac{-\sqrt[3]{x}}{h}$$

→ What if we were asked to rationalize the numerator?

$$\frac{\sqrt[3]{x+h} - \sqrt[3]{x}}{h}$$

$$\frac{-x^3}{(2x+5)^{1/2}}$$

$$\left[3x^2(2x+5)^{1/2} - x^3(2x+5)^{-1/2} \right] \cdot \frac{(2x+5)^{1/2}}{(2x+5)^{1/2}}$$

$$= \frac{3x^2(2x+5) - x^3}{(2x+5)^{3/2}}$$

$$= \frac{6x^3 + 15x^2 - x^3}{(2x+5)^{3/2}} = \frac{5x^3 + 15x^2}{(2x+5)^{3/2}}$$

$$= \frac{5x^3 + 15x^2}{(2x+5)^{3/2}} = \frac{5x^2(x+3)}{(2x+5)^{3/2}}$$